Part A: Negative Number Operations

Evaluate each expression completely.

1. $-14 - 13$
2. $-8 + (-15)$
3. $-14 - 18$
4. $19 - (-12)$
5. $30 \div (-5)$
6. $-5 \cdot 8$
7. $-6 \cdot 10$
8. $22 \div (-11) - 7$
9. $18 \div 6 - (-17)$
10. $-50 \div 5 - (-4)$

Part B: Order of Operations

Evaluate each expression completely. Show each major step.

1. $(9 + (-15))^2 \div 9$
2. $14 - 9 \cdot ((-5) + 14)$
3. $(-8 + 2)^2 \div (-6)$
4. $(-4 + (-2))^2 \div 12$
5. $(-4)^3 - 10 \cdot 10$
6. $(-2 + 10)^2 \div (-4)$
7. $-3 - 5 \cdot ((-5) + (-3))$
8. $2 - 5 \cdot (6 + 2)$
9. $(2 + (-12))^2 \div 10$
10. $20 - 6 \cdot (3 + 20)$

Part C: Square and Cube Roots

Evaluate each root.

1. $\sqrt{9}$
2. $\sqrt{25}$
3. $\sqrt{81}$

4. $\sqrt{100}$

5. $\sqrt{49}$

6. $\sqrt{4}$

7. $\sqrt[3]{64}$

8. $\sqrt[3]{8}$

9. $\sqrt[3]{27}$

10. $\sqrt[3]{125}$

Part D: Simplifying Radicals

Simplify each radical completely.

1. $\sqrt{45}$

2. $\sqrt{40}$

3. $\sqrt{90}$

4. $\sqrt{56}$

5. $\sqrt{104}$

6. $\sqrt[3]{40}$

7. $\sqrt[3]{24}$

8. $\sqrt[3]{48}$

9. $\sqrt[3]{80}$

10. $\sqrt[3]{56}$

Part E: Exponent Properties

Simplify. Exponent rules apply to multiplication, division, and powers—not addition.

1. $9x^8 \cdot 7x^4$

2. $7x^5 \cdot 7x^7$

3. $7x^5 \cdot 7x^4$

4. $\dfrac{8x^{13}}{x^8}$

5. $\dfrac{4x^{14}}{x^6}$

6. $\dfrac{5x^{16}}{x^8}$

7. $(2x^3)^3$

8. $(7x^6)^2$

9. $(7x^9)^2$

10. $(4x^4)^2$

Part F: Greatest Common Factor

Find the greatest common factor.

1. $\mathrm{GCF}\,(99, 27)$
2. $\mathrm{GCF}\,(90, 100)$
3. $\mathrm{GCF}\,(100, 30)$
4. $\mathrm{GCF}\,(50, 5)$
5. $\mathrm{GCF}\,(33, 3)$
6. $\mathrm{GCF}\left(77x^3, 99x^4\right)$
7. $\mathrm{GCF}\left(18x^4, 42x^1\right)$
8. $\mathrm{GCF}\left(15x^2, 3x^5\right)$
9. $\mathrm{GCF}\left(35x^1, 40x^2\right)$
10. $\mathrm{GCF}\left(110x^4, 99x^2\right)$

Part G: Combining Like Terms

Combine like terms. Do not add or multiply the exponents.

1. $8x - 1 + x + 8$
2. $2x + 2 - x - 5$
3. $4x - 11 + 10x + 9$
4. $-x + 12 + 2x - 7$
5. $10x - 11 + 6x + 10$
6. $2x - 6x^2 + 4x^2 + 11x$
7. $6x^2 + 11x + 3x + 10x^2$
8. $9x + 8x^2 + 4x + 3x^2$
9. $10x + 5x^2 + 8x^2 - 7x$
10. $12x^2 - 3x^2 - 6x + 12x$

Part H: Solving Linear Equations

Solve each equation for x. Show your steps.

1. $5x - 8 = -63$
2. $7x - 2 = 68$
3. $4(6x + 7) - 6 = -98$
4. $8(9x - 8) + 4 = 84$
5. $3x + 9 = 11x - 15$
6. $4x + 7 = 11x + \dfrac{77}{4}$
7. $-5x + 15 = -20x + 315$
8. $-10x + 2 = -12x + \dfrac{18}{5}$
9. $\dfrac{x - 1}{4} + 14 = \dfrac{203}{16}$
10. $\dfrac{7}{6}x + 14 = \dfrac{539}{30}$

Answer Key

Part A: Negative Number Operations

1. -27
2. -23
3. -32
4. 31
5. -6
6. -40
7. -60
8. -9
9. 20
10. -6

Part B: Order of Operations

1. 4
2. -67
3. -6
4. 3
5. -164
6. -16
7. 37
8. -38
9. 10
10. -118

Part C: Square and Cube Roots

1. 3
2. 5
3. 9
4. 10
5. 7
6. 2
7. 4
8. 2
9. 3
10. 5

Part D: Simplifying Radicals

1. $3\sqrt{5}$
2. $2\sqrt{10}$
3. $3\sqrt{10}$
4. $2\sqrt{14}$
5. $2\sqrt{26}$
6. $2\sqrt[3]{5}$
7. $2\sqrt[3]{3}$
8. $2\sqrt[3]{6}$
9. $2\sqrt[3]{10}$
10. $2\sqrt[3]{7}$

Part E: Exponent Properties

1. $63x^{12}$
2. $49x^{12}$
3. $49x^{9}$
4. $8x^{5}$
5. $4x^{8}$
6. $5x^{8}$
7. $8x^{9}$
8. $49x^{12}$
9. $49x^{18}$
10. $16x^{8}$

Part F: Greatest Common Factor

1. 9
2. 10
3. 10
4. 5
5. 3
6. $11x^3$
7. $6x$
8. $3x^2$
9. $5x$
10. $11x^2$

Part G: Combining Like Terms

1. $9x + 7$
2. $x - 3$
3. $14x - 2$
4. $x + 5$
5. $16x - 1$
6. $-2x^2 + 13x$
7. $16x^2 + 14x$
8. $11x^2 + 13x$
9. $13x^2 + 3x$
10. $9x^2 + 6x$

Part H: Solving Linear Equations

1. $x = -11$
2. $x = 10$
3. $x = -5$
4. $x = 2$
5. $x = 3$
6. $x = -\frac{7}{4}$
7. $x = 20$
8. $x = \frac{4}{5}$
9. $x = -\frac{17}{4}$
10. $x = \frac{17}{5}$